

Space Law and Policy

Applicability

Space law and policy are critical aspects due to the implications that both restrain and facilitate endeavors nations must adhere to regarding deep space travel. Moreover, these policies, principles, and laws primarily apply to international organizations and nations involved in space activities. JAXA and NASA will be among the primary organizations to which these laws apply to in the Sampling Mars Terrain to Retrieve Information and Detect Evidence of Life (STRIDE) mission as the intended objectives are to send humans to Mars to analyze the climate and geology. These policies will specifically govern nations partaking in space travel and habitat implementation and are overall legal frameworks that will govern their activities in outer space, as well as their international responsibilities and liabilities.

Policy and Principles

Promoting international cooperation on Earth is a common goal shared internationally. Working towards peaceful cooperation among humanity will help build trust among the nations, deter aggression, and deconstruct rivalries. Additionally, protecting and preserving the environment on Mars. Promoting this behavior on another planet can help humanity work towards this goal on Earth. This advancement for all humankind will benefit the overall quality of mankind, which is a NASA goal. Another result of this cooperation is that it can help improve public space awareness.

Proper contamination processes must be followed to prevent both forward and backward contamination. Planetary protection is an important principle that states that the proper sterilization of all tools and rooms needs to be carefully decontaminated and controlled. Preventing the spread of organic materials can help insure the integrity of the discovery of

extraterrestrial life. Contamination in general is impossible, but proper procedures and close monitoring must be in place to prevent any contamination.

Due to the Artemis Accords, which are legal principles agreed upon multilaterally, all legal policies and plans must be described and made transparent. Also, the Artemis Accords reaffirm the legalities previously stated in The Rescue Agreement, stating that all nations must come together to rescue the crew if a calamity arises (*Artemis Accords - NASA*, n.d.).

The science mission goals established will have the intention of energizing a competitive space market to advance the development of space technologies such as satellites, space launches, and satellite based service.

General Laws

As stated in Article II of the Outer Space Treaty (OST), no nation can claim Mars, as it is considered a celestial body. The treaty states, “Outer space, including the moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means” (NASA, 2024). In short, nobody can claim land, specific locations, a planet, or any other body in outer space. If a nation claims any land, terrain, or location on Mars, proper consequences will follow.

All activities that are conducted on the surface of Mars have to be conducted with peaceful intention. This is stated in Article 22 of the OST, with the purpose of declaring space a peaceful frontier land for humanity. NASA has also established a set of standards named Space Flight Human-System Standards that declare many standards to ensure astronaut safety and health. As stated in these standards, a safe water supply must be accessible to the astronauts on board, and on Mars. Additionally, ergonomics, habitability, and environmental health will all be considered to further follow these standards.

Japan and the U.S. are both liable for any damages they may cause to the Martian environment. If any other nation transports. These liabilities will include any damages caused by space objects on the Earth's surface or any other craft that may be damaged by the launching nation. Additionally, if any faults happen while in space, the launching nation is completely liable and will have to pay compensation.

Militarization, military installations, or military maneuvers are not allowed on the surface of Mars, as stated in the OST. Militarizing Mars will be a further step away from world peace and international cooperation. This will also result in resource competition, which can lead to the destabilization of international relations and a power imbalance (Duke, 2020). Additionally, Mars could become a property of human civilization, and militarizing this location will undermine this goal for humanity.

The safety system will be incorporated into the rocket and habitat designs. This safety system will be easily accessible for all astronauts. However, life support systems will not be required for the five cargo launches on the Starships planned for launch. But for the crew to launch on the rocket, life support systems are vital to be installed and checked before launch.

Nuclear Laws

The nuclear thermal propulsion system must pass all necessary safety inspections before being launched. A safety analysis report and an environmental analysis report must come after these safety inspections. Information regarding the consequences of a nuclear thermal launch on the environment will be included in the environmental analysis report. The project manager will compile a sufficient risk assessment analysis and a thorough mission description in the safety analysis report.

Flight considerations must be taken into account before the launch of the rocket (Pingel et al., n.d.). These considerations should include some of the most plausible accidents that could happen during transit. These flight considerations should include the rocket becoming engulfed in flame, risking nuclear pollution in the atmosphere, and the rocket crashing into the water and becoming submerged.

The development of this nuclear thermal propulsion system will be closely monitored during the construction process for the safety of all people (*Nuclear Flight Safety*, n.d.). Nuclear technicians' radiation absorption has to be closely monitored during the construction process before being distanced from the radiation source.

Japan has signed into effect The Atomic Energy Act of 1955, this law restricts the use of nuclear powers that are not for peaceful purposes. The rocket launch is considered peaceful since the rocket has not aggressive intentions, making it lawful. Additionally, the JAXA rocket launch will not be a nuclear thermal launch.

NASA will have to acquire a license issued from the Nuclear Regulatory Commission (NRC) for the nuclear-powered habitat. A Nuclear Site License will also be required for the construction and operation of the nuclear facility.

Rocket Launch Regulations

Both JAXA and NASA rockets have to pass safety inspections. These safety inspections must take into account the launch site of the rocket, since it will be launched with a nuclear thermal propulsion system.

Traffic flow on the ground must be restricted a proper amount of hours prior to the launch of the rocket. With a nuclear thermal launch, pollutants will be produced at the launch site that

can potentially cause harmful effects to bystanders. These pollutants will be kept to a minimal during the launch to reduce environmental damages.

When the rocket is launched, proper collision avoidance systems will be in place to avoid any potential collision with aircraft, or other objects.

The rocket will include specific transport methods, planned routes, and a consideration of potential hazards. These protocols are put in place to ensure crew safety in case of tragedy, or mishaps.

Satellite Laws

The satellites cannot be in such a trajectory in orbit that they risk a collision with another space object. Article 5 of the Radio Regulation Treaty states that radio frequencies must have specific transmission directions to prevent satellites from interfering with themselves (Harper, 2023). Proper tracking systems will be required to monitor their positions and trajectories to avoid collisions with other objects.

Cybersecurity laws

Proper regulations will be set in stone to protect international space infrastructures against harmful threats, which include cyberattacks. Improving resilience from any potential threats in cyberspace can ensure safety internationally.

Data encryption, access control, incident response, vulnerability scanning, penetration testing, security awareness, patch management, threat intelligence, and security audit are all necessary to be figured into the satellite(s) before the space objects deployment.

Risk Assessments Regulations

Due to the extensive risks and hazards present in the harsh space environment, limitations must be established to the extent of risk astronauts are exposed to. Regarding radiation,

astronauts are limited to a maximum amount of 600 millisieverts (mSv) for the duration of the spaceflight (in transit). Astronauts can only absorb, in a short time, 250 mSv of radiation from solar particle events (SPE). While the crew is on Mars, radiation dosage has to be closely monitored, and if the proper dose limit is approached, more extreme, proper avoidance of more radiation must be taken than what is already being done.

A regulation towards the assessment of Radiation and Risk potential is utilizing the ranking of As Low As Reasonably Achievable (ALARA). This must be utilized by NASA when appraising hazards and understanding exposure levels. Moreover, the Risk of Exposure-induced death (REID) is one that NASA must consider and communicate with management to be less than equal to 3%.

Human Factor Regulations

Food supplied to the astronauts must be appetizing to the astronaut while adhering to requirements for food safety. These foods provided must also be nutrient filled and fit the nutritional needs of each astronaut (*Space Food Systems - NASA, 2023*). When the Space Foods System is constructing the list of food that will be brought on board, astronaut's will request some of their personal favorites to be brought on board, which can be fulfilled.

Any food that is brought on board also has to be compact and lightweight. Space is limited on the rocket, and sending more goods into orbit is expensive, packing these compact foods will reduce the need for resupply and reduce total mass on board.

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